

Question:

Carlos is planting a lemon tree. The tree will cost \$90 to plant. Each year it will grow 7 lemons, which he can sell for \$1.5 each. It costs \$3 a year to water and feed the tree. How many years will it take before he starts earning money on the lemon tree?

Reasoning + Answer:

Let's think step by step :

Step 1 : Understand the costs and revenues .

- Initial cost to plant the tree : \$ 9 0
- Each year , the tree produces 7 lemons .
- Each lemon sells for \$ 1 . 5 $\rightarrow$ Revenue per year = 7 $\times$ \$ 1 . 5 = \$ 1 0 . 5 0
- Annual cost to water and feed = \$ 3

Step 2 : Calculate the net profit per year .

Net profit per year = Revenue - Water and feed cost
= \$ 1 0 . 5 0 - \$ 3 = \$ 7 . 5 0

Step 3 : We need to find out after how many years the total net profit will cover the initial \$ 9 0 cost .

So , we are solving :

Total net profit = \$ 7 . 5 0 per year $\times$ n years $\geq$ \$ 9 0

So :
 $n \geq 90 / 7.50$
 $n \geq 12$

This means that after 12 years , Carlos will have earned enough to cover the initial \$ 9 0 cost .

So , after 12 years , he will have just covered his initial cost , and from year 13 onward , he will start making a profit .

Therefore , the number of years **before** he starts earning money is **12** , but wait " starts earning money " means when he starts making a **positive** net profit , which begins the moment he covers the initial cost .

Since by the end of year 12 , his net earnings will exactly equal \$ 9 0 , and from year 13 onward he will make money (positive profit) , so the first year when he starts **earning** money is **after 12 years** , so **at year 13** .

But the question is : " How many years will it take before he starts earning money on the lemon tree ? "

We interpret this as : after how many years (i.e. , when does he start making money $\rightarrow$ i.e. , break even or after breaking even) ?

So once he has covered the cost , and gains profit , he starts earning money . That happens at the point when cumulative net profit $\geq$ \$ 9 0 .

We found : after 12 years , he has $12 \times \$ 7.50 = \$ 90$ $\rightarrow$ breaks even .

So from year 13 , he starts making **profit** (positive earnings) . So the **year** when he starts earning money is **after** 12 years .

Thus , it takes **12 years** to break even . After that , he earns money .

But the question is : " how many years will it take before he starts earning money ? "

So in common understanding , " before he starts earning money " $\rightarrow$ when does he start making a profit ?

Answer : after he has covered the cost . So at year 13 , he makes profit , so we might say it takes 13 years to **start** earning money .

Wait $\rightarrow$ but let's double-check : In year 1 , he earns \$ 1 0 . 5 0 , spends \$ 3 , so net \$ 7 . 5 0 profit $\rightarrow$ not enough to cover \$ 9 0 .

After 12 years :
Total net profit = $12 \times 7.50 = 90$ $\rightarrow$ breaks even .

So at the end of year 12 , his total profit = \$ 9 0 , which just covers the cost .

So at year 13 , he starts **earning** money $\rightarrow$ in the sense that his net earnings are positive and exceed the cost .

Therefore , the number of years until he starts earning money is **13** ?

Wait $\rightarrow$ no . He starts earning money when he has spent \$ 9 0 , and earns more than cost $\rightarrow$ but that happens after year 12 . But the question is " how many years will it take before he starts earning money " ?

At the end of year 12 , he has broken even . So from year 13 onwards , he earns profit . So strictly , " he starts earning money " at